Defining and Collecting Data

Chapter 1: 50 recall cards

101 pages: cover + question and answer pages

Recall the answer, then advance one page.
Companion to the supplied chapter guide.

Foundations

Why does data quality come first?

Foundations

Poor definitions, biased samples, and inaccurate measurements undermine conclusions, even with advanced analysis.

Recall the distinction and explain why it matters.

DCOVA

What are the five DCOVA tasks?

DCOVA

Define, Collect, Organize, Visualize, Analyze. Establish the business objective before applying these tasks.

Recall the distinction and explain why it matters.

Operational definitions

What makes a variable operationally defined?

Operational definitions

Specify exactly what it measures, the observational unit, units, time period, and inclusion rules.

Recall the distinction and explain why it matters.

Variables

What is a variable?

Variables

A characteristic that takes different values across observations. An observational unit is the person or item measured.

Recall the distinction and explain why it matters.

Variable types

How do categorical and numerical variables differ?

Variable types

Categorical values identify groups. Numerical values express quantities for which arithmetic has a meaningful interpretation.

Recall the distinction and explain why it matters.

Discrete and continuous

Counts or measurements?

Discrete and continuous

Discrete variables have countable values, such as purchases. Continuous variables can take any value in an interval, such as elapsed time.

Recall the distinction and explain why it matters.

Recording choices

Can the same concept have different variable types?

Recording choices

Yes. Age in years is numerical. Age groups are categorical. Classification depends on the operational definition.

Recall the distinction and explain why it matters.

Nominal scale

What information does a nominal scale provide?

Nominal scale

Categories without an inherent order. Examples: payment method and product category. Numeric codes remain labels.

Recall the distinction and explain why it matters.

Ordinal scale

Why are satisfaction ratings ordinal?

Ordinal scale

Categories have an order, but equal distances between adjacent categories are not established.

Recall the distinction and explain why it matters.

Interval scale

What distinguishes an interval scale?

Interval scale

Equal differences are meaningful, but zero is arbitrary. A 30 C temperature is 10 C above 20 C, without being 1.5 times as hot.

Recall the distinction and explain why it matters.

Ratio scale

When are ratios meaningful?

Ratio scale

With equal intervals and a meaningful zero. A 20-minute duration is twice a 10-minute duration.

Recall the distinction and explain why it matters.

Measurement overview

How do the four scales build on one another?

Measurement overview

Nominal: categories. Ordinal: order too. Interval: meaningful differences too. Ratio: meaningful ratios too.

Recall the distinction and explain why it matters.

Method choice

Does measurement scale alone determine the analysis?

Method choice

No. Also consider the objective, research design, sampling structure, distribution, and assumptions.

Recall the distinction and explain why it matters.

Primary and secondary

What distinguishes primary from secondary data?

Primary and secondary

Primary data are collected for the current purpose. Secondary data were collected previously for another purpose. The distinction depends on the user and purpose.

Recall the distinction and explain why it matters.

Data sources

What five data sources does the guide identify?

Data sources

Distributed sources, designed experiments, surveys, observational studies, and ongoing business activities.

Recall the distinction and explain why it matters.

Experiments

Why does random assignment matter?

Experiments

It helps create comparable treatment groups and supports causal inference when the experiment is properly designed.

Recall the distinction and explain why it matters.

Observation and surveys

What can observation and surveys establish?

Observation and surveys

Observation records behavior without assigned treatments. Surveys collect reported responses. Association alone does not establish causation.

Recall the distinction and explain why it matters.

Population and sample

How do a population and sample differ?

Population and sample

The population is the full group of interest. A sample is a selected subset used to learn about that group.

Recall the distinction and explain why it matters.

Parameter and statistic

Which describes a population, and which a sample?

Parameter and statistic

A parameter describes the population. A statistic summarizes the sample and may estimate a parameter under suitable design and assumptions.

Recall the distinction and explain why it matters.

Census

Does a census eliminate every source of error?

Census

A complete census has no sampling error. Coverage, nonresponse, and measurement errors may still occur.

Recall the distinction and explain why it matters.

Data structure

What are structured, unstructured, and semistructured data?

Data structure

Structured: fixed rows and fields. Unstructured: free-form text or media. Semistructured: some organization with flexible content.

Recall the distinction and explain why it matters.

Data cleaning

What should an analyst check before analysis?

Data cleaning

Missing values, impossible values, invalid categories, inconsistent formats, duplicate records, and unusual observations. Document decisions.

Recall the distinction and explain why it matters.

Outliers

Should an unusual observation always be deleted?

Outliers

No. Investigate whether it is an error or a valid extreme value. Preserve the raw record and document any correction or exclusion.

Recall the distinction and explain why it matters.

Recoding

What makes grouped categories valid?

Recoding

Mutually exclusive: no overlap. Collectively exhaustive: every eligible value fits. State boundaries and missing-value handling.

Recall the distinction and explain why it matters.

Sampling frame

What is a sampling frame?

Sampling frame

The list or mechanism from which units are selected. Compare it with the target population to detect omissions or ineligible entries.

Recall the distinction and explain why it matters.

Probability sampling

What defines a probability sample?

Probability sampling

Each population unit has a known, positive selection probability under the design. Probabilities need not be equal.

Recall the distinction and explain why it matters.

Nonprobability sampling

How do convenience and judgment samples differ?

Nonprobability sampling

Convenience uses accessible participants. Judgment uses researcher expertise. Unknown selection probabilities limit design-based population inference.

Recall the distinction and explain why it matters.

Simple random sampling

What defines a simple random sample of fixed size?

Simple random sampling

Every possible sample of that size has the same selection probability. A complete frame is usually needed.

Recall the distinction and explain why it matters.

Replacement

What changes when sampling with replacement?

Replacement

A selected unit can appear again. Without replacement, selected units cannot be selected a second time.

Recall the distinction and explain why it matters.

Systematic sampling

How would you sample 200 of 10,000 items?

Systematic sampling

For this divisible frame, k = N/n = 50. Choose a random start from 1 to 50, then select every 50th item.

Recall the distinction and explain why it matters.

Systematic limitations

What threatens a systematic sample?

Systematic limitations

Periodicity aligned with the sampling interval can produce an unrepresentative sample. Check the ordering before selection.

Recall the distinction and explain why it matters.

Stratified sampling

How does stratified sampling select units?

Stratified sampling

Divide the population into nonoverlapping strata and randomly sample within every stratum. This supports subgroup representation.

Recall the distinction and explain why it matters.

Sampling allocation

What if a small stratum is oversampled?

Sampling allocation

Its sample share exceeds its population share. Use design-appropriate weights when estimating overall population quantities.

Recall the distinction and explain why it matters.

Cluster sampling

How does one-stage cluster sampling work?

Cluster sampling

Randomly select some natural groups, then observe all eligible units in those groups. It can reduce travel and fieldwork costs.

Recall the distinction and explain why it matters.

Two-stage sampling

What is two-stage cluster sampling?

Two-stage sampling

First randomly select clusters. Then randomly sample units within the selected clusters.

Recall the distinction and explain why it matters.

Strata and clusters

What is the central distinction?

Strata and clusters

Stratified: sample units from every group. Cluster: select some groups. Similar units within clusters can reduce precision.

Recall the distinction and explain why it matters.

Survey errors

What four survey errors should you assess?

Survey errors

Coverage, nonresponse, sampling, and measurement. Several can occur in the same survey.

Recall the distinction and explain why it matters.

Coverage

What is coverage error?

Coverage

The frame fails to match the target population, for example when a survey of all students uses a full-time-only list.

Recall the distinction and explain why it matters.

Nonresponse

When can nonresponse create bias?

Nonresponse

When missing respondents differ systematically on the quantity of interest. A response rate alone does not determine the size of the bias.

Recall the distinction and explain why it matters.

Sampling error

Will increasing sample size remove bias?

Sampling error

Larger well-designed samples usually reduce random sampling variation. They do not automatically remove coverage, nonresponse, or measurement bias.

Recall the distinction and explain why it matters.

Measurement error

What question features can distort responses?

Measurement error

Leading wording, ambiguity, double-barreled questions, unclear reference periods, recall difficulties, and interviewer influence.

Recall the distinction and explain why it matters.

Neutral wording

How can a leading satisfaction question be improved?

Neutral wording

Ask how satisfied or dissatisfied the respondent was with the service during the most recent visit. Avoid praising the service in the question.

Recall the distinction and explain why it matters.

Survey ethics

What does honest survey reporting require?

Survey ethics

Disclose the target population, frame, selection method, sample size, response limitations, wording, and uncertainty. Avoid manipulating selection or interpretation.

Recall the distinction and explain why it matters.

Literary Digest

What is the key lesson of the 1936 poll example?

Literary Digest

A very large response count cannot guarantee representativeness. Frame coverage and differential nonresponse both matter.

Recall the distinction and explain why it matters.

GT&M acquisition

What evidence supports an acquisition decision?

GT&M acquisition

Consistently defined sales and expenses, economic context, and relevant customer data. Extra profitability and risk measures are teaching extensions, not a calculated valuation.

Recall the distinction and explain why it matters.

Whitney Wireless

How should a best-sales-year claim be checked?

Whitney Wireless

Request comparable annual revenue and unit sales across all relevant years, with clear scales, accounting definitions, and reporting periods.

Recall the distinction and explain why it matters.

Ashland MultiComm

Does falling renewal prove prices are too high?

Ashland MultiComm

No. Examine subscriber counts, revenue, and renewal rates. Price, competition, service, and streaming are possible explanations to investigate.

Recall the distinction and explain why it matters.

CardioGood Fitness

How should its customer variables be classified?

CardioGood Fitness

Model: nominal. Self-rated fitness 1-5: ordinal. Income and age: numerical. Weekly sessions: discrete counts. Distance: continuous measurement.

Recall the distinction and explain why it matters.

Clear Mountain surveys

Can 62 undergraduate and 44 graduate responses prove representativeness?

Clear Mountain surveys

No. Assess selection, frame, and nonresponse. The separate questionnaires and datasets were not supplied, so their exact variables cannot be confirmed.

Recall the distinction and explain why it matters.

Integrated review

A frame omits part-time students, 400 of 1,000 reply, and wording is leading. Which errors apply?

Integrated review

Coverage from omitted students. Potential nonresponse bias from missing replies. Sampling variation from selection. Measurement error from leading wording.

Recall the distinction and explain why it matters.